Violet Monserate

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Computer Engineering & Philosophy student passionate about hardware, robotics, accessibility, and security. Experienced in collaborative development across software and CAD, with a strong background in K-12 STEM education.

Skills

Hard Skills: Java, C/C++ (Arduino, ESP-IDF, STM32), C# (.NET Core, LINQ, Entity Framework), Verilog, SQL, Azure Kubernetes Service, Oracle (ORDS, PL/SQL), ROS, JS/TS (React, Astro), HTML/CSS, Git, SolidWorks, Altium
Soft Skills: problem-solver, detail-oriented, creative, adaptive, fast learner, concise documentation, time management

Education

University of Washington, Paul G. Allen School of CSE, M.S. in Computer Science & Engineering — Expected *06/28*
University of Washington, B.S. in Computer Engineering & B.A. in Philosophy: Ethics | **3.97 GPA** — 09/23 - *06/27*
CS: Data Structures, Algorithms, Systems Programming, Digital Circuits and Systems, Hardware/Software Design, Computer Security, Cryptography, Compilers, Databases, HCI, Autonomous Robotics
Philosophy: Computer Ethics, Intergenerational Ethics, Neuroethics, Philosophy of Science, Philosophy for Children

Projects

Assorted Firmware for Custom Circuits- C++, Arduino, STM32, ESP-IDF, GIT | UW Superbike — *09/25 - current*
- Develop CAN main board using ESP-IDF for ESP32 while maintaining original functionality from Teensy 2.0
- Generate and debug LVBMS system on STM32, communicating with ICs over I2C and other boards over CAN
- Map out and design lightweight UI to display metrics (e.g. speed, voltage) using LGVL receiving data over CAN

Assistive Museum Headphones- Python, C++, STM32 Ecosystem, Git | UW Undergrad Capstone — *10/23 – 06/25*
- Develop Python CV pipeline on Raspberry Pi 4 to detect fiducial markers and trigger real-time audio
- Program STM32F4 firmware using FreeRTOS to manage I2S audio streaming and UART button controls

Rover Remote Control UI- JavaScript, React, Vite, Cesium, C++, Git | Husky Robotics Team — *10/23 – 06/25*
- Write React widget to track position, heading, and path of rover against a 3-D topographical map with Cesium
- Develop back-end server to deliver glTF tiles with a RESTful URI to operate without an internet connection
- Improve rover controls, telemetry, and pathing, updated and delivered through React Redux

Mobile Manipulator- CAD, Onshape, ROS2, Python | WEIRD Lab, Paul G. Allen School — 03/24-09/24
- Designed physical mobile platform utilizing a holonomic drive base and two 6-DOF robot arms

Experience

Teaching Assistant- Paul G. Allen School of Computer Science & Engineering | Seattle, WA — *08/24 – current*
- Create lesson plans to concepts in C, assembly, CS theory, and social impact through lecture/section activities
- Support 1-on-1 in office hours to understand course content & foster problem-solving and collaborative skills

Mentor, Lead Mentor- Changemakers in Computing | Seattle, WA — *01/24 - 08/24, 03/26-current*
- Direct program operations and train new mentors through task delegation and staff onboarding
- Spearhead web dev and tech ethics curriculum for 40 students, leading to 97% program recommendation rate

Software Development Intern- INIT SE | Seattle, WA — *06/25 –09/25*
- Developed RESTful GET/POST methods for Oracle Database and client onboarding utilizing OAuth 2.0
- Created an Automatic Passenger Counter demo using CAD, documenting details and BOM using Office 365
- Implemented a log parser in C#, utilizing IoC through Dependency Injection (DI) with database extraction and email notifications, enhancing debugging and efficiency

STEM Alternative Spring Break Instructor- University of Washington CELE | Seattle, WA — *01/25 - 05/25*
- Developed ~5 hours' worth of culturally competent STEM education for 6th-8th grade students with peers
- Taught and built relationships with ~60 students for 1 week in a rural community in Northeast Washington

Coding Instructor- Coding with Kids | Redmond, WA — *06/23 - 08/24*
- Designed curriculum for >450 primary school students and integrated coding into interdisciplinary projects
- Maintained 90% retention, and positive feedback from 95% of parents and guardians

Scholar- Apple & Kode with Klossy | New York, NY — *06/22 - 08/22*
- Learned best practices for AI and Machine Learning to create app that identifies and sorts recycling
- Presented the AI app to an audience of >400 in Apple 5th Ave @ NYC, including AI/ML engineers